MD. ASIF HASAN

AI & Systems Researcher | *East West University, Dhaka, Bangladesh*

Dhaka, Bangladesh • asifhasan099@gmail.com • (+880) 1550032232

asif.info.bd • github.com/asicoltd • linkedin.com/in/asif-hasan-099

Research Interests

Core Domains: Spatio-Temporal Graph Neural Networks (TGNNs, Dynamic Graphs), Self-Supervised & Contrastive Representation Learning (MoCo v2, SimCLR), Streaming Network Telemetry & Dynamic Anomaly Detection, Causal Evaluation under Distribution Shift, Efficient Deep Learning Inference (Quantization, Model Optimization).

Education

East West University **Dhaka, Bangladesh**

Master of Science in Computer Science & Engineering ***(CGPA: 3.56 / 4.00)*** *Dec 2024 – Oct 2026*

- **M.Sc. Thesis (Completed):** *Momentum-Contrastive Spatio-Temporal Graph Attention Networks for IoT Intrusion Detection (MC-STGAT-IDS)*.
- **Supervisor:** Dr. Ahmed Wasif Reza (Professor & Dean, Faculty of Sciences and Engineering).
- **Academic Trajectory:** Marked upward progression from undergraduate studies; focused on Advanced Graph Learning and Continuous Streaming Analytics.

East West University **Dhaka, Bangladesh**

Bachelor of Science in Computer Science & Engineering ***(CGPA: 3.08 / 4.00)*** *Jan 2020 – Dec 2024*

- **Undergraduate Thesis:** *Delay Tolerant Network (DTN) Routing Optimization with Machine Learning*.
- **Focus:** Evaluated predictive forwarding and opportunistic routing using the ONE simulator under sparse mobility; coupled with empirical radio propagation and wireless node telemetry (HAM Radio Callsign: **S21NEO**).

Publications & Manuscripts

Peer-Reviewed Conference Proceedings:

[1] **Md. Asif Hasan**, M. H. Azhar, M. A. Sarkar, A. H. Esha, M. A. Morshed, and A. W. Reza. "Ensemble Deep Learning for Enhanced Detection and Classification of Hand, Foot, and Mouth Disease." In *Proceedings of the International Conference on Data Mining and Information Security (ICDMIS 2024)*, *Lecture Notes in Networks and Systems (LNNS)*, vol. 1388, pp. 359–376. Springer, Singapore, 2025.

[2] P. Islam, **Md. Asif Hasan**, I. H. Farabi, S. M. S. Sobhani, K. M. S. Kamal, and A. W. Reza. "Development of a Deep Learning Model for Accurate Classification of Peptic Ulcer and Digestive Tract Disorders Using Endoscopic Images." In *ICDMIS 2024*, *LNNS*, vol. 1389, pp. 449–461. Springer, Singapore, 2025.

[3] P. Islam, Z. D. Ziden, S. Abdullah, **Md. Asif Hasan**, M. A. Asif, M. Khan, and T. Mittra. "Rose Disease Detection Using YOLO: A Deep Learning-Based Approach." In *Intelligent Strategies for ICT*, *LNNS*, vol. 1381, pp. 147–159. Springer, Singapore, 2025.

Manuscripts in Preparation:

[4] **Md. Asif Hasan** and A. W. Reza. "Momentum-Contrastive Spatio-Temporal Graph Attention for Streaming Network Telemetry." *Manuscript detailing M.Sc. thesis architecture and forward-chronological causal evaluation; preparing for preprint/submission*.

Professional & Research Experience

Financial Reporting Council (FRC), Ministry of Finance, Bangladesh **Dhaka, Bangladesh**

AI Intern (AI Engineering & Systems) *Nov 2025 – Jun 2026*

- **LLM Regulatory Auditing Pipelines:** Engineered end-to-end open-source LLM compliance analysis workflows for automated auditing of corporate disclosures against statutory IFRS/IAS benchmarks.
- **Production RAG Architecture:** Designed multi-stage Retrieval-Augmented Generation pipelines integrating PaddleOCR for multi-page financial filings, dense vector embeddings (FAISS), and LLM reasoning, achieving a 25% improvement in context retrieval accuracy.
- **Low-Latency Model Quantization:** Optimized 4-bit and 8-bit quantized models (AWQ, GGUF via `llama.cpp`) for DeepSeek-R1 and Qwen2.5 on GPU memory-constrained servers, reducing inference latency by 40% while preserving assessment fidelity.
- **Deterministic Schema Validation:** Implemented structured prompt templates producing 100% schema-validated JSON outputs, eliminating hallucinations across 500+ corporate financial filings.

Selected Research Projects

MC-STGAT-IDS: Momentum-Contrastive Spatio-Temporal GAT for IoT Security **Jan 2026 – Oct 2026**

Technologies: PyTorch Geometric, Dynamic TGNNs, MoCo v2, Network Telemetry

- Built an inductive spatio-temporal graph attention architecture in PyTorch Geometric processing continuous telemetry streams without discrete windowing artifacts.
- Integrated causal temporal masking with momentum contrast (MoCo v2) under strict forward-chronological evaluation, preventing temporal data leakage and combating distribution collapse across dynamic evolving topologies.
- Designed multi-head attention interpretability matrices to attribute root-cause anomaly nodes in dynamic intrusion graphs.

Self-Supervised Contrastive Representation Learning for Medical Imaging **Apr 2026 – Jun 2026**

Technologies: SimCLR, MoCo v2, BYOL, PyTorch, ResNet, Data Augmentation

- Benchmarked self-supervised visual representation frameworks (SimCLR, MoCo v2, BYOL) using ResNet backbones on MRI data.
- Demonstrated 92% diagnostic classification accuracy under extreme low-label regimes ($< 10\%$ labeled data), reducing annotation dependency by 40% compared to standard supervised baselines.

Hybrid Deep Neural Recommender Engine **Jan 2026 – Mar 2026**

Technologies: TensorFlow, Scikit-learn, BERT Embeddings, Microsoft MIND Benchmark

- Engineered a hybrid neural recommender combining dense text embeddings (BERT) and collaborative user-item interaction matrices.
- Evaluated ranking accuracy on the Microsoft MIND benchmark, achieving a 23% improvement in user engagement over baseline models.

Multi-Modal Financial Market Forecasting & Reinforcement Learning **Jul 2025 – Oct 2025**

Technologies: FinBERT Sentiment Extraction, Temporal LSTM, PPO Trading Agent, PyTorch

- Built a multi-modal forecasting framework combining fine-tuned FinBERT sentiment extraction with LSTM temporal regressors.
- Trained a Proximal Policy Optimization (PPO) reinforcement learning agent on adaptive multi-asset allocation, achieving 15% improved risk-adjusted returns over baseline market strategies.

Technical & Algorithmic Skills

- **Algorithmic Problem Solving:** Solved **800+ algorithmic problems** on Codeforces, HackerRank, and Beecrowd (Graph Traversal/Flows, Dynamic Programming, Tree Decomposition, Algorithmic Complexity). Regular hackathon competitor.
- **Graph & Deep Learning Architectures:** Spatio-Temporal Graph Neural Networks (GAT, STGCN, TGNNs), Contrastive & Self-Supervised Learning (MoCo v2, SimCLR), Large Language Models (Quantization, RAG, Structured Output), Time-Series Modeling (TFT, LSTM).
- **Frameworks & Libraries:** PyTorch, PyTorch Geometric (PyG), DGL, HuggingFace Transformers, LangChain, FAISS, OpenCV, Scikit-learn, Pandas, NumPy.
- **Languages & Developer Systems:** Python, C++, SQL, Java, Git, Docker, Linux/WSL, CUDA, `llama.cpp`, LaTeX.

Honors, Leadership & Community Service

- **Licensed Amateur Radio Operator (Callsign: S21NEO):** Officially licensed by Bangladesh Telecommunication Regulatory Commission (BTRC). Conducted RF antenna propagation and wireless telemetry experiments; contributed emergency communications during disaster relief.
- **Scouting & Humanitarian Leadership (10+ Years):** Active Scout Leader (2026–Present); Senior Rover Mate (2019–2025), Samatat Open Scout Group. Awarded the **Messenger of Peace Ring Badge** by WOSM and the **National Service Award**.
- **Civic & Healthcare Volunteer Service:** Completed **25+ voluntary blood donations** through the Quantum Foundation. Former Treasurer (2019–2020), Leo Club of Dhaka Imperial, directing community welfare fundraisers and health campaigns.
- **Competitive Debating: Champion**, EWUDC Intra-University Debate Championship (Bangla Segment, 2023). Former General Secretary & Trainer, College Debating Club (2018–2019).
- **Athletic Discipline & Endurance:** Completed **20+ marathons** organized by Run Bangladesh (and co-organized 4 endurance events). Holds Blue Belt in Karate and Green Belt in Taekwondo.
- **Language Proficiency:** English (IELTS Academic 7.0; CEFR C1 Level), Bengali (Native).

Academic References

Dr. Ahmed Wasif Reza
Professor & Dean, Faculty of Sciences and Engineering
East West University, Dhaka, Bangladesh
wasif@ewubd.edu (+880) 1780099173

Dr. Mohammad Rifat Ahmmad Rashid
Associate Professor, Department of Computer Science & Engineering
East West University, Dhaka, Bangladesh
rifat.rashid@ewubd.edu (+880) 1314447616